II Semester Syllabus

Course Code: CSPC201	Data Structures	Credits 3-0-0: 3

Syllabus:

Unit I	Introduction to data structures: definition, types (linear and non-linear). Definition, memory representation, types of arrays (one-dimensional, two-dimensional, multi-dimensional). Operations on arrays: insertion, deletion, traversal, searching, and sorting (basic introduction). Introduction to linked lists, memory representation, types of linked lists (singly linked list, doubly linked list, circular linked list). Operations on linked lists: insertion, deletion, traversal, and reversal of linked lists. Comparison between arrays and linked lists.	8 lectures
Unit II	Stacks: Definition, array and linked list implementations of stacks. Operations: push, pop, and peek. Applications of stacks: recursion, infix to postfix conversion, postfix evaluation, and checking balanced parentheses in expressions. Queues: Definition, array and linked list implementations of queues. Types of queues: simple queue, circular queue, and priority queue. Operations on queues: enqueue, dequeue, and peek. Applications of queues in task scheduling and buffering. Recursion: Introduction to recursion, difference between recursion and iteration. Types of recursion: tail recursion, head recursion. Examples of recursive algorithms: factorial, Fibonacci series, tower of Hanoi, and backtracking algorithms.	9 lectures
Unit III	**Trees:** Definition, terminology (node, edge, height, depth, etc.). Types of trees: binary trees, binary search trees (BST), AVL trees, and expression trees. Tree traversal techniques: in-order, pre-order, post-order traversal. Operations on binary trees: insertion, deletion, searching. **Binary Search Trees (BST):** Definition, properties, operations (insertion, deletion, searching). Balanced trees: introduction to AVL trees and rotations. **Heaps:** Definition, types of heaps (min-heaps and max-heaps), heap properties. Operations on heaps: insertion, deletion, heapify. Priority queues and their applications.	9 lectures

Unit IV	**Graphs:** Introduction to graphs, terminology (vertices, edges, degree, etc.). Representation of graphs: adjacency matrix, adjacency list. Types of graphs: directed and undirected graphs, weighted and unweighted graphs. **Graph Algorithms:** Depth-First Search (DFS), Breadth-First Search (BFS), shortest path algorithms (Dijkstra's algorithm, Bellman-Ford algorithm). Minimum spanning trees: Prim's and Kruskal's algorithms. **Algorithm Analysis:** Introduction to algorithm analysis, time and space complexity. Asymptotic notation: Big-O, Big-Theta, Big-Omega. Best, average, and worst-case analysis.	10 lectures
Unit V	**Hashing:** Introduction to hashing, hash functions, properties of good hash functions. Collision resolution techniques: chaining, open addressing (linear probing, quadratic probing, double hashing). Applications of hashing in data retrieval and database indexing. **Sorting Algorithms:** Basic sorting algorithms: Bubble sort, Selection sort, Insertion sort. Advanced sorting algorithms: Merge sort, Quick sort, Heap sort. Comparison of sorting algorithms in terms of time and space complexity.	6 lectures

Course Code: CSPC202	**Data Structures Lab**	**Credits 0-0-4: 2**

List of Programs:

1. Write a program to insert, delete, and traverse elements in a one-dimensional array.
2. Implement a program to perform matrix operations (addition, subtraction, multiplication) using two-dimensional arrays.
3. Write a program to insert, delete, and reverse elements in a singly linked list.
4. Create a program to implement a circular linked list and demonstrate insertion and deletion at different positions.
5. Write a program to implement a doubly linked list and perform operations like insertion, deletion, and traversal.
6. Develop a program to implement a stack using arrays and perform push, pop, and peek operations.
7. Write a program to implement a queue using linked lists and perform enqueue and dequeue operations.
8. Implement a program to convert an infix expression to postfix using a stack.
9. Write a recursive program to solve the Tower of Hanoi problem.
10. Create a recursive program to calculate the nth Fibonacci number.
11. Write a program to implement a binary search tree and perform insertion, deletion, and searching.
12. Develop a program to perform in-order, pre-order, and post-order traversal of a binary tree.

13. Write a program to implement a min-heap and perform insertion and deletion operations.
14. Create a program to implement an AVL tree and perform rotations to maintain balance after insertions.
15. Write a program to implement a priority queue using a max-heap.
16. Write a program to represent a graph using an adjacency matrix and perform Depth-First Search (DFS).
17. Develop a program to represent a graph using an adjacency list and perform Breadth-First Search (BFS).
18. Write a program to implement Dijkstra's algorithm to find the shortest path from a source to all other vertices in a weighted graph.
19. Create a program to find the Minimum Spanning Tree of a graph using Prim's algorithm.
20. Write a program to implement Kruskal's algorithm for finding the Minimum Spanning Tree of a graph.
21. Write a program to implement linear probing for collision resolution in a hash table.
22. Create a program to implement quadratic probing and demonstrate insertion and searching in a hash table.
23. Write a program to implement bubble sort and analyze its time complexity.
24. Develop a program to implement quick sort and compare its performance with merge sort.
25. Write a program to implement heap sort and display the sorted elements of an array.

Course Code: PHBS201	**Physics For Engineers**	**Credits 3-0-0: 3**

Unit I	Wave Mechanics and X-ray Diffraction: Wave-particle duality, de-Broglie matter waves, Phase and Group velocities, Davisson-Germer experiment, Heisenberg uncertainty principle and its applications, Wave function, and its significance, Schrödinger's wave equation – particle in a one-dimensional box. Diffraction of X-rays by crystal planes, Bragg's spectrometer, Compton's effect.	10 lectures
Unit II	Dielectric and Magnetic Properties of Materials: Dielectric constant and Polarization of dielectric materials, Types of Polarization (Polarizability). Equation of internal fields in liquid and solid (One-Dimensional), Clausius Mussoti- Equation, Ferro and Piezo electricity (Qualitative), Frequency dependence of dielectric constant, Dielectric Losses, Important applications of dielectric material, Langevin's theory for dia and paramagnetic material, Phenomena of hysteresis and its applications. Ultrasonic: Generation, detection, and application of ultrasonics.	8 lectures
Unit III	Electromagnetics: Displacement Current, Maxwell's Equations (Integral and Differential Forms). Equation of continuity, EM- Wave equation and its	6 lectures

	propagation characteristics in free space and in conducting media, Poynting theorem and Poynting vectors.	
Unit IV	Modern Optics: Laser: Spontaneous and stimulated emission of radiation, population inversion, the concept of 3 and 4 level Laser, construction and working of Ruby, He-Ne lasers, and laser applications. Fiber Optics: Fundamental ideas about optical fiber, Propagation mechanism, Acceptance angle and cone, Numerical aperture, Propagation Mechanism and communication in fiber Single and Multi-Mode Fibers, step index, and graded index fiber. Holography: Basic Principle of Holography, Construction, and reconstruction of Image on hologram and applications of holography.	9 lectures
Unit V	Semiconducting Materials: Concept of energy bands in solids, Carrier concentration and conductivity in intrinsic semiconductors and their temperature dependence, carrier concentration, and conductivity in extrinsic semiconductors and their temperature dependence. Hall effect in semiconductors, Compound semiconductors, and Optoelectronic Materials. Superconducting Materials: Temperature dependence of resistivity in superconducting materials, Effect of magnetic field (Meissner effect), Temperature dependence of critical field, Type I and Type II superconductors. Nano-Materials: Basic principle of nanoscience and technology, structure, properties, and uses of Fullerene and Carbon nanotubes, Applications of nanotechnology.	9 lectures

Course Code: PHBS202	**Physics For Engineers Lab**	**Credits 0-0-2: 1**

List of Experiments:

1. To study the variation of the magnetic field along the axis of the current-carrying circular coil.
2. To study the Hall's effect and to determine the Hall coefficient in n-type Germanium.
3. To study the energy band gap of n-type Germanium using the four-probe method.
4. To draw the hysteresis curve of a given sample of ferromagnetic material.
5. Determination of wavelength of laser source by diffraction rating method.
6. NA of optical fiber.

Course Code: MABS201	**Mathematics – II**	**Credits 3-0-0 : 3**

Syllabus:

Unit I	Linear differential equation of higher order with constant and variable coefficients, method of variation of parameters, Cauchy Euler's and Legendre's equations, series solution of ordinary differential equations with special emphasis on Legendre's and Bessel's differential equations; application of linear differential equation of second order.	10 lectures
Unit II	Fourier Series: Periodic function, Fourier series of a function with 2π period and arbitrary period, Fourier series of even and odd functions, half range Fourier sine and cosine series.	8 lectures
Unit III	Laplace transform and its properties; existence theorem; Laplace transform of derivatives, inverse Laplace transform; convolution theorem, use of Laplace transform in solving differential equations.	8 lectures
Unit IV	Fourier integral, Fourier transform, Fourier sine and cosine transforms and their elementary properties, applications of Fourier transform in boundary value problems**.**	8 lectures
Unit V	Introduction to Z-transform and its convergence, Properties of Z-transform. Inverse Z-transform by Partial Fraction Method, Power Series Expansion, Convolution theorem.	8 lectures

Course Code: CSES201	**Programming for Problem Solving**	**Credits 2-0-0: 2**

Syllabus:

Unit I	Introduction to computer systems, programming environments, and programming languages. Basic concepts of algorithms, flowcharts, and pseudo-code. Introduction to Python: History, features, structure of Python programs. Installing Python and setting up an IDE. Understanding Python syntax, indentation, variables, data types, and operators. Input and output in Python (input(), print()). **Problem-Solving Aspect:** Writing basic Python programs for arithmetic operations. Demonstrating simple problems using flowcharts and algorithms.	7 lectures
Unit II	Operators in Python (arithmetic, relational, logical, assignment, bitwise, and membership operators). Operator precedence and associativity. Decision statements: if, if-else, if-elif-else. Loop control statements: for, while, break, continue, pass. Functions: Defining functions, function parameters (positional, keyword, default, variable-length arguments), return values, recursion. **Problem-Solving Aspect:** Implementing decision-making and looping programs, such as a simple calculator, grading system, and pattern printing.	7 lectures
Unit III	Built-in Data Structures: Lists: Creation, indexing, slicing, list methods. Tuples: Immutable sequences. Dictionaries: Key-value pairs, dictionary operations. Sets: Unique unordered collections. Introduction to Object-	8 lectures

	Oriented Programming (OOP): Classes and objects. Constructors (__init__ method). Instance and class variables. Inheritance and polymorphism. **Problem-Solving Aspect:** Writing programs using lists, dictionaries, and sets. Implementing a simple Student Record System using classes.	
Unit IV	Lambda functions and list comprehensions. Exception handling (try, except, finally).n File handling in Python: Working with text and binary files (open(), read(), write()). Handling file operations (append, seek, tell). Exception handling in file operations. **Problem-Solving Aspect:** Implementing programs for file-based data storage. Writing programs for error handling and automation.	7 lectures
Unit V	Introduction to NumPy and Pandas for data handling. Matplotlib for data visualization. Basics of web scraping using BeautifulSoup. Introduction to APIs and JSON data handling. Automating tasks using Python scripts. **Problem-Solving Aspect:** Creating simple data analysis projects using Pandas. Visualizing data using Matplotlib. Writing a web scraper to extract data from a website.	7 lectures

Course code: CSES202	**Programming for Problem Solving Lab**	**Credits 0–0–4: 2**

List of Experiments:

1. Write a Python script that demonstrates the use of variables, keywords, and different data types (numeric, string, list, dictionary, Boolean, tuple, set). Include creation, deletion, and access operations)
2. Write a program (script) for conversion of temperature measured in degree Celsius to Fahrenheit. Use the 4 different ways of printing the results. Create two versions of the program, in which you get the value of the temperature in Celcius: i) manually ii) using command-line parameters.
3. Create a program that asks the user for their age and prints different messages using if, if-else, and nested if-else statements. For example, determine if the user is a minor, adult, or senior.
4. Write a program that prints the first 10 numbers of the Fibonacci sequence using both a for loop and a while loop. Include control statements (break, continue, pass) in different scenarios.
5. Write a script which asks for the details of a student of NERIST such as, name, DoB, module, year, state, gender, etc. Assign the student to the hostel number calculated as $(x\%n+1)$, where x is the sum of date, month and year of the DoB of the student and n is the number of hostels. Print the details of the student along with the hostel number using the different ways of printing.

6. Write a program that sums a sequence of integers supplied as command line parameters.[Hint: To get the number of arguments supplied, you can use len(argv).
7. Develop a function that takes variable-length arguments (e.g., using *args or **kwargs) and returns their sum or concatenation. Create test cases to demonstrate its functionality.
8. Define a function called hypotenuse that calculates the length of the hypotenuse of a right triangle when the two sides are given. Use this function in a program to determine the length of the hypotenuse of a triangle. The function should take two arguments and return the hypotenuse.
9. Write a function that takes the time as three integer arguments (for hours, minutes, and seconds), and returns the number of seconds since the last time the clock "struck 12". Use this function to calculate the amount of time in seconds between two times, both of which are within one 12-hour cycle of the clock.
10. Write a recursive function power (base, exponent) that when invoked returns base exponent
11. Write a Python function swap(i,j,l) which interchanges the ith and jth elements in the list, l.
12. Write a Python program which generates a list of the first n whole numbers with its corresponding factorials. The list will look like [[0,1], [1,1], [2, 2], [3,6], [4, 24]….].
13. A tuple object is immutable. Hence, an item in a tuple cannot be modified. Also, an item cannot be added. However, a workaround is to convert a tuple to a list using the list() function and modify the list and convert back to a tuple using the tuple() function.
14. Write the following Python functions and demonstrate how they can be used in the main module:

    a) appendTuple(T,x) which returns a tuple as a result. The result is tuple, T appended with item x as the last item.

    b) insertTuple(T,i,x) which returns a tuple as a result. The result is tuple, T into which item x is inserted before the ith index.

    c) ModifyTuple(T,i,x) which returns a tuple as a result. The result is tuple, T whose item in the ith index is overwritten by x.

    d) PopTuple(T) which returns a tuple as a result. The first item of the returned tuple is the last item in T and the second item is tuple T from which the last item is remove.

15. Create a vector with values ranging from 10 to 49
16. Use the NumPy module to create a 2D matrix. Perform the following operations: creation, deletion, access, and manipulation (e.g., transpose, reshape). Display the matrix before and after each operation.
17. Write a Python program that calculates the determinant, inverse, rank, and eigenvalues of a given matrix using NumPy. Test the program with a sample 2x2 matrix.
18. Implement a program that uses NumPy to solve a system of linear equations. Provide an example of three equations with three variables and output the solution.
19. Create a Python script that generates a bar graph and a pie chart based on sample data (e.g., sales data). Label the axes and include titles for each plot.
20. Develop a program that generates a histogram of a dataset and a scatter plot showing the relationship between two variables. Use Matplotlib for visualization and provide appropriate titles and labels.
21. Design a program that takes user input for a series of numbers, processes this data to

calculate the mean and standard deviation, and then displays the results in a histogram and a line plot. Use NumPy for calculations and Matplotlib for plotting.

22. Create and manipulate DataFrames using Pandas. Perform the following :
    i) Access, filter, and modify data within a DataFrame.
    ii) statistical analysis and handle missing data.
    iii) Group data and export DataFrames to CSV files.
    iv) Visualize data using Matplotlib.

Course code: CSES204	Computing Workshop	Credits 0–0–4: 2

Syllabus:

Unit I	Unix (Linux) basics: shell, file system, permissions; Unix (Linux) commands; Introduction to Shell Scripting (Bash): looping, conditional statements, creation of new commands using shell scripting, automating jobs with Cron Jobs and shell scripts. Working with text data (grep, awk, sed).
Unit II	**Introduction to R/Python for Data Analysis:** Basics of R/Python syntax, Data structures, Handling structured and unstructured data, Statistical Methods & Probability: Descriptive statistics: Mean, Median, Mode, Variance, Standard Deviation. Probability distributions. **Data Analysis & Visualization:** Visualizing distributions, correlations, and trends.
Unit III	**Shell Scripting & Scientific Computing:** Scientific Computing with Scilab: Matrix operations, numerical solutions, visualization in Scilab.
Unit IV	**Applications in Text, Sequence & Image Processing** Text Processing & Natural Language Processing (NLP): Tokenization, stemming, lemmatization. Named Entity Recognition (NER), Sentiment Analysis. Introduction to Image Processing & Computer Vision: Image enhancement, filtering, edge detection (OpenCV, Scikit-Image).
Unit V	**Embedded Systems & IoT with Raspberry Pi** Interfacing Raspberry Pi with Sensors & Actuators: GPIO programming, data collection, real-time processing. IoT Data Logging & Visualization: Cloud integration, remote monitoring, edge computing. Mini-Project: End-to-End Application Development: Weather monitoring, Smart automation, Image-based alert system, etc.

Course Code: UHVA201	Universal Human Values	Credits 2-1-0: 3

Syllabus :

Unit I	**Introduction to Value Education:** Right understanding, relationship and physical facility (Holistic development and the role of education); Understanding value education; Self-exploration as the process for value education; Continuous happiness and prosperity – the basic human aspirations; Happiness and prosperity – Current scenario; Method to fulfill the basic human aspirations.	6 lectures
Unit II	**Harmony in the Human Being:** Understanding human being as the co-existence of the self and the body; Distinguishing between the needs of the self and the body; The body as an instrument of the self; Understanding harmony in the self; Harmony of the self with the body; Programme to ensure self-regulation and health.	6 lectures
Unit III	**Harmony in the Family and Society:** Harmony in the family – the basic unit of human interaction; Trust – the foundational value in relationship; Respect – as the right evaluation; Other feelings, justice in human-to-human relationship; Understanding harmony in the society; Vision for the universal human order.	6 lectures
Unit IV	**Harmony in the Nature/Existence:** Understanding harmony in the nature; Interconnectedness, self-regulation and mutual fulfilment among the four orders of nature; Realizing existence as co-existence at all levels; The holistic perception of harmony in existence.	4 lectures
Unit V	**Implications of the Holistic Understanding – A look at Professional Ethics:** Natural acceptance of human values; Definitiveness of (Ethical) human conduct; A basis for humanistic education, humanistic constitution and universal human order; Competence in professional ethics; Holistic technologies, production systems and management models-typical case studies; Strategies for transition towards value-based life and profession.	6 lectures